

HTML5

HTML



What is new in HTML5?

- HTML5 provides new tags and attributes
- HTML5 introduces new tags for better **semantic description of content** (e.g., `<nav>`, `<header>`, `<footer>`, ...), for direct inclusion of multimedia elements (`<audio>` and `<video>`), and for graphic rendering powered by JavaScript (`<canvas>`)
- HTML5 drops some deprecated elements from HTML 4.01 (e.g., `<font>` and `<center>`)

HTML5 vs. XHTML2

XHTML2

- It could have become the current version of HTML (instead of HTML5)
- Several new features
 - e.g., *href* attribute in any tag (e.g., `<li href="">...</li>`),
 - any element can be an image (e.g., `<p src="im.jpg"> </p>`)
- Special emphasis on XML technologies (e.g., *XForms*)
- Total separation between content and presentation

HTML5

- New tags and attributes
- Less "strict" than XHTML2

W3C Official Recommendation...

<https://www.w3.org/TR/2014/REC-html5-20141028/>



HTML5

A vocabulary and associated APIs for HTML and XHTML

W3C Recommendation 28 October 2014

This Version:

<http://www.w3.org/TR/2014/REC-html5-20141028/>

Latest Published Version:

<http://www.w3.org/TR/html5/>

Latest Version of HTML:

<http://www.w3.org/TR/html/>

Latest Editor's Draft of HTML:

<http://www.w3.org/html/wg/drafts/html/master/>

Previous Version:

<http://www.w3.org/TR/2014/PR-html5-20140916/>

Previous Recommendation:

<http://www.w3.org/TR/1999/REC-html401-19991224/>

... W3C Official Recommendation...

<https://www.w3.org/TR/html52/>



HTML 5.2

W3C Recommendation, 14 December 2017

This version:

<https://www.w3.org/TR/2017/REC-html52-20171214/>

Latest published version:

<https://www.w3.org/TR/html52/>

Latest published version of HTML:

<https://www.w3.org/TR/html/>

Previous version:

<https://www.w3.org/TR/2017/PR-html52-20171102/>

Editor's Draft:

<https://w3c.github.io/html/>

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W3C Recommendation

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... W3C Official Recommendation

<https://www.w3.org/TR/html53/>

W3C Working Draft

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HTML 5.3

W3C Working Draft, 18 October 2018



This version:

<https://www.w3.org/TR/2018/WD-html53-20181018/>

Latest published version:

<https://www.w3.org/TR/html53/>

Editor's Draft:

<https://w3c.github.io/html/>

Previous Version:

<https://www.w3.org/TR/2018/WD-html53-20180809/>

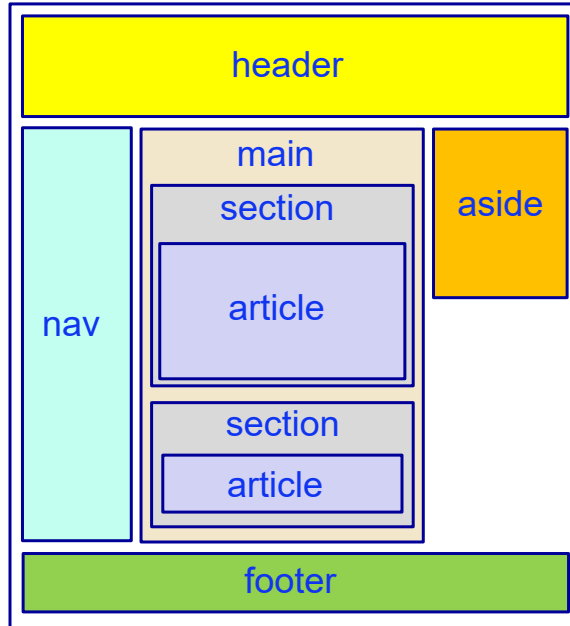
Test Suite:

<https://github.com/web-platform-tests/wpt/tree/master/html>

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Structural elements of HTML5



`<header>`

“introductory” information

`<main>`

main content of the page

`<nav>`

navigation elements

`<section>`

generic section (similar to `<div>`)

`<article>`

“independent” content portion

`<aside>`

“extra” content

`<footer>`

“bottom area” of a content region

Audio

Three supported formats

Browser	MP3	WAV	OGG
Edge/IE	YES	YES*	YES*
Chrome	YES	YES	YES
Firefox	YES	YES	YES
Safari	YES	YES	NO
Opera	YES	YES	YES

*From Edge 79

<audio> tag

```
<audio src="mysong.mp3" controls>
```

Sorry, your browser doesn't support the 'audio' tag

```
</audio>
```


Video

Three supported formats

Browser	MP4	WebM	Ogg
Edge	YES	YES	YES
Chrome	YES	YES	YES
Firefox	YES	YES	YES
Safari	YES	YES	NO
Opera	YES	YES	YES

<video> tag

```
<video width="320" height="240" controls>  
  <source src="video.ogv" type="video/ogg">  
  Sorry, your browser doesn't support the 'video' tag  
</video>
```

WebM and Ogg video

WebM

- Is a free, open, media file format designed for the Web (<https://www.webmproject.org>)
- *WebM* files consist of video streams compressed with the *VP8* video codec (created by Google – better quality and performance compared to H.264) and audio streams compressed with the *Vorbis* audio codec

Ogg

- Is a free, open container format maintained by the *Xiph.Org* Foundation
- *Ogg* files consist of video streams compressed with the *Theora* codec (derived from the formerly proprietary VP3 codec)

Canvas

Exploits JavaScript to draw on the page



```
<html>
  <body>
    <canvas id="mioCanvas" width="200" height="100"
      style="border:1px solid #c3c3c3;">
      Sorry, your browser doesn't support the
      'canvas' tag
    </canvas>
    <script type="text/javascript">
      var c = document.getElementById("mioCanvas");
      var cxt = c.getContext("2d");
      cxt.fillStyle = "#FF0000";
      cxt.beginPath();
      cxt.arc(70,18,15,0,Math.PI*2,true);
      cxt.closePath();
      cxt.fill();
    </script>
  </body>
</html>
```

New types for forms

color, date, datetime, datetime-local, email, month, number, range, search, tel, time, url, week

Examples:

Homepage: `<input type="url" id="user_url">`

Points: `<input type="number" id="points" min="1" max="10">`

Birthday: `<input type="date" id="user_date">`

Birthday:

novembre 2015

lun	mar	mer	gio	ven	sab	dom
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6